

Q.21 Explain measurement of glass viscosity.

Q.22 Explain the etching technique of decoration.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Differentiate between borosilicate glass and soda-lime glass based on composition and properties.

Q.24 Discuss various processes of glass decoration.

Q.25 Discuss refractories used in various parts of glass tank furnace.

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5th Sem / Ceramic

Subject : Glass Technology - II

Time : 3 Hrs.

SECTION-A

M.M. : 60

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Primary component of commercial glass is _____

- a) Alumina (Al_2O_3)
- b) Silica (SiO_2)
- c) Boron trioxide (B_2O_3)
- d) Calcium oxide (CaO)

Q.2 The typical temperature range for annealing soda-lime glass is _____

- a) 200-300°C b) 500-600°C
- c) 800 - 900 °C d) 1000 - 1200°C

Q.3 Which of these oxide is commonly used to produce a blue color in glass?

- a) Cobalt (Co) b) Chromium (Cr)
- c) Selenium (Se) d) Manganese (Mn)

(20)

(4)

220454

(1)

220454

Q.4 Increase in the temperature of glass will _____
viscosity of molten glass.

- a) Increases
- b) decreases
- c) remains constant
- d) fluctuates unpredictably

Q.5 Which of these glass is used in optical lenses due to its high refractive index?

- a) Soda-lime glass b) Lead glass
- c) Fused silica glass d) Borosilicate glass

Q.6 Among these furnaces which type of furnace is used in glass manufacturing for melting?

- a) Blast furnace b) Open-hearth furnace
- c) Glass tank furnace d) Cupola furnace

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 _____ oxide is primarily responsible for the green tint in glass. (Iron oxide / Cobalt oxide)

Q.8 Lehr is used for _____ of glass. (annealing / toughening)

(2)

220454

Q.9 _____ used to remove the greenish tint caused by iron impurities in glass. (MnO_2/UO_2)

Q.10 Zirconia refractories are preferred for the sidewalls & bottom of a glass tank furnace. (T/F)

Q.11 Viscosity of glass _____ as it approaches to the melting point? (increases/decreases)

Q.12 Sulfuric acid is commonly used for glass etching? (T/F)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Discuss the meaning of supercooled liquid in the context of glass formation?

Q.14 Explain testing of glass density.

Q.15 Discuss importance of glass viscosity during melting.

Q.16 Name the colorants of glass.

Q.17 Explain importance of solar energy.

Q.18 Write sources of air, water and noise pollution in Glass industry.

Q.19 Explain the process of tempering of glass.

Q.20 Explain theory of decolorization and give two decoloring agents.

(3)

220454